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HEADPHONES AND CARRYING CASE

Abstract

To provide headphones with an improved noise canceling function and the like. Headphones include a headband, a housing, an ear pad attached to the housing, a hanger that supports the housing, and a slider that connects the hanger to an end portion of the headband. The headphones further include a microphone arranged at a place of the housing outside a movable range of the hanger. The housing includes a case that covers a movable range of the hanger. The hanger and the microphone are arranged in the same space formed in the housing on an outer peripheral side of the hanger.

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Background/Summary

TECHNICAL FIELD

[0001] A technology disclosed in the present specification (hereinafter referred to as “present disclosure”) relates to headphones and a carrying case for headphones.

BACKGROUND ART

[0002] Headphones are widely known as acoustic equipment combining devices that convert an electrical signal output from a reproduction device or a receiver into sound waves (audible sound) using a sounding body (speaker or the like) close to the ear (eardrum). Recently, with the spread of compact and high-performance reproduction devices, the use of headphones is also increasing.

[0003] Headphones are often used outdoors, on the road, on a moving train, or the like, and noise resistance is strongly desired. Therefore, high-performance and high-definition headphones (see Patent Document 1, for example) having a noise canceling function that cancels noise in an external environment and provides the listener with a good music reproduction environment have been developed and are already used.

[0004] In addition, in order to facilitate carrying of the headphones and to store the headphones and preventing damage due to external force at the time of storage, a carrying case for the headphones has also been developed (see Patent Document 2, for example), and is often supplied together with the headphones or separately.

[0005] In addition, the pad portion of the headphones that comes into contact with the ear is made of a material such as polyurethane excellent in cushioning property, but there is a problem that the pad portion is hydrolyzed under the influence of perspiration and body temperature to deteriorate over time.

[0006] For this reason, there are cases where the ear pad is replaceable (see Patent Document 3, for example), and a replacement pad at the time of deterioration is commercially available. In addition, headphones including an ear pad having no air permeability in consideration of sound insulation have been proposed (see Patent Document 4, for example).

CITATION LIST

Patent Document

[0007] Patent Document 1: Japanese Patent Application Laid-Open No. 2008-116782 [0008] Patent Document 2: Japanese Patent Application Laid-Open No. 2013-38671 [0009] Patent Document 3: Japanese Patent Application Laid-Open No. 2010-187151 [0010] Patent Document 4: Japanese Patent Application Laid-Open No. 2021-164085 [0011] Patent Document 5: Japanese Patent Application Laid-Open No. 2010-62733 [0012] Patent Document 6: Japanese Patent Application Laid-Open No. 2006-344813

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

[0013] An object of the present disclosure is to provide headphones and a carrying case for headphones with an improved noise canceling function and the like.

Solutions to Problems

[0014] The present disclosure has been made in view of the above problems, and a first aspect thereof is headphones including [0015] a headband, a housing, an ear pad attached to the housing, a hanger that supports the housing, and a slider that connects the hanger to an end portion of the headband, in which [0016] the headphones further include a microphone arranged at a place of the

housing outside a movable range of the hanger.

[0017] The housing includes a case that covers a movable range of the hanger, and has a slit for operating a joint portion of the hanger with the head band. Moreover, the hanger and the microphone are arranged in the same space formed in the housing. For example, the microphone is arranged on an outer peripheral side of the hanger.

[0018] Moreover, the slider has a seamless short-arc hollow pipe shape, and contains a cable connected to a circuit in the housing in the hollow pipe. Specifically, the slider includes a resin having a seamless short-arc hollow pipe shape and a shaft twist slider including a metal nut component inserted in a joint portion between the resin and the hanger, and the slider rotatably supports the hanger about a center line of the slider by a fastening structure between the metal nut component and a distal end of a shaft twist inserted into the hanger.

[0019] Furthermore, the ear pad is configured by containing a ring-shaped urethane foam in a skin, and the skin is configured by bonding a thin urethane film integrated with at least one of a bottom surface or a front surface of a wet foam layer having a three-dimensional microporous structure to a base cloth. Alternatively, the ear pad is configured by containing a ring-shaped urethane foam in a skin, and the skin is configured by bonding a wet foam layer having a three-dimensional microporous structure to a front surface of a product obtained by bonding an air-impermeable moisture-permeable waterproof film to any one of a bottom surface and a front surface of a wet base cloth having open cells or air permeability.

[0020] In addition, the ear pad includes a fixing claw including a finger hook portion provided on the inner peripheral side of the ring shape. The fixing claw has a claw fitting structure with a claw fixing portion on the housing side. The ear pad includes fixing claws arranged at a plurality of locations on an inner periphery. The fixing claws arranged at a plurality of locations include a fixing claw including a finger hook portion and a fixing claw not including a finger hook portion. The fixing claws including a finger hook portion are arranged side by side in a range where a curvature radius R of the ear pad is small at the front of the face of the wearer.

[0021] In addition, a wiring portion including a hole for inserting a wire rod is provided in a wall separating a space in the housing, and acoustic sealing of the space in the housing is achieved by inserting a bushing into the hole so as to press the wire rod against a wall surface of the hole. The bushing has a rib formed over the entire circumference, and the rib interferes with a wall surface of the hole when the bushing is inserted into the hole.

[0022] Furthermore, a second aspect of the present disclosure is a carrying case for the headphones mainly according to the first aspect of the present disclosure, including: [0023] a top surface lid portion; [0024] a bottom surface tray portion; and [0025] a side surface portion having a ring shape sewn to the top surface lid portion and the bottom surface tray portion and having a fold that is folded by a pressing force in a vertical direction.

Effects of the Invention

[0026] According to the present disclosure, it is possible to provide headphones and a carrying case for headphones with an improved noise canceling function and the like.

[0027] Note that the effects described in the present specification are merely examples, and the effects brought by the present disclosure are not limited thereto. Furthermore, the present disclosure may further provide additional effects in addition to the effects described above.

[0028] Still other objects, features, and advantages of the present disclosure will become apparent from a more detailed description based on embodiments as described later and the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0029] FIG. **1** is a diagram illustrating the appearance of headphones **100**.

[0030] FIG. **2** is a diagram illustrating the appearance (state in which slide function is used) of the headphones **100**.

[0031] FIG. **3** is a diagram illustrating a functional configuration of a noise canceling system.

[0032] FIG. **4** is a four-view diagram of the headphones **100** according to an embodiment of the present disclosure.

[0033] FIG. **5** is a perspective view of a housing **101**.

[0034] FIG. **6** is a cross-sectional view of the housing **101** taken along line A-A of FIG. **4**.

[0035] FIG. **7** is a side view and a cross-sectional view of the housing **101** (when tilt angle of hanger **103** is minimum).

[0036] FIG. **8** is a side view and a cross-sectional view of the housing **101** (when tilt angle of hanger **103** is maximum).

[0037] FIG. **9** is a diagram illustrating an arrangement example of microphones in a case where the hanger **103** is Y-shaped and has two movable shafts.

[0038] FIG. **10** is a diagram illustrating an arrangement example of microphones in a case where the hanger **103** has one movable shaft.

[0039] FIG. **11** is a diagram illustrating an arrangement example of microphones in a case where the hanger **103** has one movable shaft.

[0040] FIG. **12** is a diagram collectively illustrating microphone arrangement examples of a plurality of patterns.

[0041] FIG. **13** is a diagram illustrating an arrangement example in which four microphones are arranged separately in a first floor portion and a second floor portion of the housing **101**.

[0042] FIG. **14** is an enlarged view of a joint portion between a slider **104** and a hanger **103** in the headphones **100** according to the present disclosure.

[0043] FIG. **15** is a diagram illustrating a cross section of a joint portion between the slider **104** and the hanger **103**.

[0044] FIG. **16** is a diagram illustrating the vicinity of the lower end of the slider **104** into which a shaft twist slider **1503** is inserted.

[0045] FIG. **17** is a perspective view of the slider **104**.

[0046] FIG. **18** is a side view of the slider **104**.

[0047] FIG. **19** is a cross-sectional view of the slider **104**.

[0048] FIG. **20** is an enlarged view of a joint portion between the slider **104** and the hanger **103**.

[0049] FIG. **21** is a diagram illustrating a state in which air flows around the vicinity of the joint portion between the slider **104** and the hanger **103**.

[0050] FIG. **22** is a plan view of **22** an ear pad **107**.

[0051] FIG. **23** is a cross-sectional view of the ear pad **107** taken along line B-B.

[0052] FIG. **24(A)** is a diagram illustrating a cross-sectional structure of a normal wet synthetic leather, and FIG. **24(B)** is a diagram illustrating a cross-sectional structure of a normal dry synthetic leather.

[0053] FIG. **25** is a diagram illustrating a cross-sectional structure of an example of synthetic leather used for a skin containing urethane foam.

[0054] FIG. **26** is a diagram illustrating a modification of the synthetic leather illustrated in FIG. **25**.

[0055] FIG. **27** is a diagram illustrating another modification of the synthetic leather illustrated in FIG. **25**.

[0056] FIG. **28** is a diagram illustrating a cross-sectional structure of another example of synthetic leather used for a skin containing urethane foam.

[0057] FIG. **29** is a diagram illustrating a modification of the synthetic leather illustrated in FIG.

28.

[0058] FIG. **30** is a diagram illustrating a cross-sectional structure of yet another example of synthetic leather used for a skin containing urethane foam.

[0059] FIG. **31** is a diagram illustrating a modification of the synthetic leather illustrated in FIG. **30**.

[0060] FIG. **32** is a diagram illustrating a cross-sectional structure of still another example of synthetic leather used for a skin containing urethane foam.

[0061] FIG. **33** is a diagram illustrating a modification of the synthetic leather illustrated in FIG. **32**.

[0062] FIG. **34** is a diagram illustrating a cross-sectional view of the ear pad **107** and the housing **101**.

[0063] FIG. **35** is a further enlarged view of a fixing claw on the ear pad **107** side and a claw fixing portion on the housing **101** side in the cross-sectional view illustrated in FIG. **34**.

[0064] FIG. **36** is a diagram illustrating an example in which six claw fitting structures are arranged in the ear pad **107** and the housing **101**.

[0065] FIG. **37** is a perspective view illustrating arrangement places of two fixing claws arranged side by side in a range where R is small.

[0066] FIG. **38** is a diagram illustrating a state in which a finger is inserted into one of the two fixing claws illustrated in FIG. **37** to remove the fixing claw.

[0067] FIG. **39** is a diagram illustrating a state in which an ear (auricle) is inserted into an elliptical shape of the ear pad **107**.

[0068] FIG. **40** is a diagram illustrating a modification of the finger hook portion of the fixing claw.

[0069] FIG. **41** is a diagram illustrating another modification of the finger hook portion of the fixing claw.

[0070] FIG. **42** is a diagram illustrating another example in which the plurality of claw fixing portions is arranged.

[0071] FIG. **43** is a diagram illustrating a state in which a cushion material is wrapped around a wire rod.

[0072] FIG. **44** is a diagram illustrating an external configuration example of a bushing **4400**.

[0073] FIG. **45** is a diagram illustrating a state (cross-sectional view) in which the bushing **4400** is inserted into a component **4500** forming a hole.

[0074] FIG. **46** is a diagram illustrating a state (cross-sectional view) in which the bushing **4400** is inserted into the component **4500** forming the hole.

[0075] FIG. **47** is a diagram illustrating an influence of attachment of the bushing **4400** on frequency characteristics of the headphones **100**.

[0076] FIG. **48** is a diagram illustrating a modification of the arrangement place of the wiring portion (or place where bushing is applied) in the case of the housing **101**.

[0077] FIG. **49** is a diagram illustrating another modification of the arrangement place of the wiring portion (or place where bushing is applied) in the housing **101**.

[0078] FIG. **50** is a six-view diagram of the appearance of a carrying case **5000** as viewed from upper, lower, front, rear, left, and right sides.

[0079] FIG. **51** is a two-view diagram in which the carrying case **5000** is opened as viewed from the upper side and from one side surface.

[0080] FIG. **52** is an exploded view of the carrying case **5000** disassembled into its main components.

[0081] FIG. **53** is a diagram illustrating a developed view of an upper gusset portion **5021** and a lower gusset portion **5022**.

[0082] FIG. **54** is a diagram illustrating a state in which a side surface portion of the carrying case **5000** is folded.

[0083] FIG. **55** is a diagram illustrating a part where a fold is formed on a side surface portion of

the carrying case **5000**.

[0084] FIG. **56** is a diagram illustrating a part where a fold is formed on a side surface portion of the carrying case **5000**.

[0085] FIG. **57** is an enlarged view of the vicinity of the upper end of the hanger **103**.

[0086] FIG. **58** is a diagram illustrating the vicinity of the lower end of the slider **104** into which a shaft twist slider **1503** is inserted.

[0087] FIG. **59** is a diagram illustrating a basic structure of headphones.

[0088] FIG. **60** is a diagram illustrating a state in which a step is generated between a slider and a hanger when the hanger is twisted.

MODE FOR CARRYING OUT THE INVENTION

[0089] Hereinafter, the present disclosure will be described in the following order with reference to the drawings.

[0090] A. Basic structure of headphones

[0091] B. Noise canceling function

[0092] C. Optimal arrangement of noise canceling microphones

[0093] D. Slider structure

[0094] E. Ear pad structure

[0095] E-1. Moisture permeable synthetic leather ear pad structure

[0096] E-2. Replaceable claw ear pad

[0097] F. Wiring portion sealing structure

[0098] G. Carrying case

A. Basic Structure of Headphones

[0099] First, a basic structure of headphones will be described with reference to FIG. **59**. Note, however, that FIG. **59** is merely used for convenience to describe basic components of headphones for reference, and the present disclosure is not limited thereto.

[0100] The illustrated headphones **5900** are a so-called overhead type, and include a pair of left and right housings **101**, a headband **102** that clips the pair of left and right housings **101** to the head of the wearer (not illustrated), and a pair of left and right hangers **103** that couple the pair of left and right housings **101** with left and right end portions of the headband **102**. The pair of left and right hangers **103** are fixed to left and right end portions of the headband **102** via sliders **104**, respectively.

[0101] A speaker **106** and an ear pad **107** are arranged in each of the pair of left and right housings **101**. The speaker **106** includes a sounding body such as a diaphragm (described later). A surface of the housing **101** from which the sound from the speaker **106** is output is also referred to as an “acoustic surface” for convenience. The acoustic surface is basically a side surface facing the ear of the wearer in the worn state. The ear pad **107** is attached to the acoustic surface and forms a noise canceling (NC) region in which its sound insulation property is secured. In the case of the housing **101**, a printed wiring board on which electric components including a drive circuit of the speaker **106** are mounted is incorporated.

[0102] The headband **102** is curved in an arc shape and provided with elasticity, and is clipped to the head using the elastic force to press the pair of left and right housings **101** against the left and right ears of the wearer and hold the headphones **5900** on the head of the wearer. Note that the headband **102** may have a shape in which the curvature of the arc is curved gently to soften the feeling of pressing to the ear.

[0103] The lower end of a slider **104** is coupled to the upper end of the hanger **103**. The method for coupling the slider **104** and the hanger **103** is not particularly limited (method for coupling slider **104** and hanger in present disclosure will be described later). Additionally, the upper end of the slider **104** is inserted into the headband **102** from an end portion of the hollow headband **102**, and is attached so as to be stretchable in an extending direction of the headband **102** indicated by reference numeral **112**. Therefore, by sliding the slider **104** in the extending direction **112** of the

headband **102**, the length of the slider **104** can be adjusted to fit the individual wearer's head size such that the housing **101** (acoustic surface on which speaker **106** is located) is just at the position of the ear.

[0104] Moreover, the hanger **103** is attached via the slider **104** so as to be rotatable about a center line **111** in an extending direction **112** of the headband **102** as a rotation axis. In short, the slider **104** has a slide function of sliding the position of the hanger **103** in the extending direction **112** and also a twist function of rotating the hanger **103** about the center line **111**. Therefore, in a state where the headphones **5900** are worn on the head of the wearer, the hanger **103** rotates about the center line **111** of the headband **102**, and the acoustic surface of the housing **101** has a rotation angle just following the shape of the temporal region of the wearer's head (or inclination of ear).

[0105] The hanger **103** is formed in a Y shape. The upper end of the hanger **103** is attached to the lower end of the head band **102** via the slider **104**. Additionally, the housing **101** is attached to the bifurcated lower end of the hanger **103**. Here, the housing **101** is supported by the hanger **103** so as to be rotatable about a broken line **113** connecting the bifurcated lower ends of the hanger **103** as a rotation axis. Therefore, in a state where the headphones **5900** are worn on the head of the wearer, the housing rotates about the rotation axis **113**, and the acoustic surface of the housing **101** has a tilt angle just following the shape of the temporal region (or inclination of ear) different for each wearer.

[0106] FIG. **1** illustrates the appearance of headphones **100** according to the best embodiment to which the present disclosure is applied. Similarly to the headphones **5900** exemplified for reference in FIG. **59**, the headphones **100** include a pair of left and right housings **101** for holding speakers **106**, a headband **102** that clips the pair of left and right housings **101** to the head of the wearer (not illustrated), and a pair of left and right hangers **103** that couple the pair of left and right housings **101** with left and right end portions of the headband **102**. The pair of left and right hangers **103** are fixed to the left and right end portions of the headband **102** via sliders **104** that stretch and contract in the longitudinal direction.

[0107] The headphones **100** illustrated in FIG. **1** are the same in that the headband **102** has a function of being clipped to the head using elastic force to wear the headphones **100** on the head of the wearer, the slider **104** has a slide function of sliding the position of the hanger **103** and a twist function of rotating the hanger **103** about the center line of the slider **104**, and the hanger **103** has a function of adjusting the tilt angle of the housing **101**. For reference, FIG. **2** illustrates a state in which the hanger **103** is extended from the end portion of the headband **102** by the slide function of the slider **104** on the near side (note, however, that illustration of the other housing **101** and hanger **103** is omitted).

[0108] The headphones **100** have many differences from the headphones **5900** illustrated in FIG. **59**, such as that the slider **104** has a cylindrical shape and that the hanger **103** (at least bifurcated portion of Y shape and rotation shaft portion supporting housing **101**) is arranged in the case of the housing **101**, but main differences serving as features of the present disclosure will be described later.

B. Noise Canceling Function

[0109] In order to enjoy music without being disturbed by external sound, high performance and high quality headphones having a noise canceling function have been developed and are already used. Noise canceling is generally achieved by a combination of passive noise canceling and active noise canceling. Passive noise canceling is achieved by maintaining the sound insulation property of a noise canceling (NC) region with an ear pad or the like. On the other hand, active noise canceling is a technology of canceling the low range to the middle range of external sound by generating a sound having an opposite phase to the external sound to cancel the external sound.

[0110] Active noise canceling includes a "feedforward method" that cancels an external sound in an NC region using a signal having an opposite phase to a signal picked up by a microphone (hereinafter also referred to as "feedforward (FF) microphone") installed outside the NC region,

and a “feedback method” that cancels an external sound in the NC region using a signal having an opposite phase to a signal picked up by a microphone (hereinafter also referred to as “feedback (FB) microphone”) installed in the NC region.

[0111] FIG. 3 schematically illustrates a functional configuration of a noise canceling system in which the feedforward method and the feedback method are combined. An NC region **301** is a target region where noise canceling is performed, and corresponds to a space between the ear of the wearer wearing the headphones and the case (housing) of the headphones. The ear canal entrance is shielded by an ear pad or the like. The illustrated noise canceling system includes a feedforward (FF) microphone **302**, an FF cancellation signal generation unit **303**, a reproduction unit **304**, a synthesis unit **305**, a feedback (FB) microphone **306**, and an FB cancellation signal unit **307**.

[0112] The FF microphone **302** is installed outside the NC region **301**, and attempts to pick up an external sound generated outside the NC region **301**. Then, the FF cancellation signal generation unit **303** analyzes a signal picked up by the FF microphone **302** and generates an FF cancellation signal having an opposite phase to cancel the analyzed external sound. The reproduction unit **304** includes an acoustic element such as a speaker installed in the NC region **301**, and audio-outputs an FF cancellation signal in the NC region **301**. Note that in a case where the headphones are reproducing music, the FF cancellation signal is synthesized with a music signal in the synthesis unit **305**, and the reproduction unit **304** outputs an audio signal after synthesizing the music signal and the FF cancellation signal. With this arrangement, the external sound leaking into the NC region **301** and the FF cancellation signal reproduced from the reproduction unit **304** cancel each other, and it is difficult to hear the external sound in the NC region **301**. In a case where the headphones are reproducing music, the external sound is canceled in the NC region **301**, and it is easy to listen to the music.

[0113] Furthermore, the FB microphone **306** is installed inside the NC region **301**. It is difficult to completely cancel the external sound leaking into the NC region **301** only by the FF cancellation signal, and there is a case where the external sound is left uncanceled. The FB microphone **306** picks up the residual sound left uncanceled. Then, the FB cancellation signal generation unit **307** analyzes a signal picked up by the FB microphone **306**, and generates an FB cancellation signal having an opposite phase to cancel the analyzed residual sound. The FB cancellation signal is synthesized with the FF cancellation signal in the synthesis unit **305**. In a case where the headphones are reproducing music, the FB cancellation signal is synthesized with the music signal together with the FF cancellation signal in the synthesis unit **305**. The reproduction unit **304** outputs an audio signal obtained by synthesizing the FB cancellation signal and the FF cancellation signal. With this arrangement, the external sound left uncanceled by the FF cancellation signal and the FB cancellation signal cancel each other, and it is further difficult to hear the external sound in the NC region **301**. Therefore, in a case where the headphones are reproducing music, the external sound is canceled by the FF cancellation signal and the external sound left uncanceled is canceled by the FB cancellation signal, so that it is easier to listen to music in the NC region **101**. That is, accuracy of noise canceling can be further enhanced by combining the FB method with the FF method.

[0114] Hereinafter, simply referring to “noise canceling” in the present specification basically refers to active noise canceling of the feedforward method. Therefore, the noise canceling microphone in the embodiment described later is arranged not in the NC region partitioned by the ear pad but outside the NC region such as the housing.

C. Optimal Arrangement of Noise Canceling Microphones

[0115] As described in the above section A, the hanger that adjusts the tilt angle of the housing is often incorporated in the housing so that the headphones can be correctly worn according to the shape of the temporal region different for each wearer.

[0116] Meanwhile, in order to achieve the noise canceling function and other various types of signal processing on the basis of sound information outside the headphones, headphones equipped

with a plurality of microphones for noise canceling are used. In order to obtain good sound “quality” and “directivity” by performing signal processing on sound signals picked up by a plurality of microphones, it is important to arrange microphones close to the mouth of the wearer, secure a large distance between microphones, and take measures against wind noise, for example. [0117] In the case of a structure in which the hanger **103** is attached to the outermost peripheral portion outside the case of the housing **101** as in the headphones **5900** illustrated in FIG. **59**, in order to be able to pick up sound without being affected by a change in the tilt angle of the hanger **103**, the arrangement of microphones is limited to the inside of the case of the housing on the inner peripheral side of the hanger **103**, and it is difficult to secure a large inter-microphone arrangement. For example, when the hanger **103** exists near the front of the opening of the microphone, a change in the tilt angle of the hanger **103** in the wearing state different for each wearer affects sound information that can be picked up by the microphone, and there is also a problem that turbulence generated by air flowing around the hanger **103** and the case of the housing **101** during walking of the wearer becomes wind noise and is picked up by the microphone.

[0118] In addition, when the arrangement of the microphones is limited to the inner peripheral side of the hanger **103**, there are many cases where the microphones have to be arranged in a space contributing to the acoustic circuit due to the movement of the hanger **103** (change in tilt angle or the like). In this case, not only the sound generated from the headphones **101** themselves (or speaker **106**) becomes noise, but also a complicated sealed structure around the microphone is required, which makes it difficult to design and manufacture the housing.

[0119] In short, when the hanger **103** operates outside the case of the housing **101**, the influence on the space in front of the microphone hole increases. For example, when the tilt angle of the hanger **103** becomes small and the hanger **103** approaches the front of the microphone hole, the air flows around the hanger **103** and the case of the housing **101** to easily generate turbulence, which causes wind noise to the microphone.

[0120] Therefore, in the present disclosure, first, the microphone is arranged on the outer peripheral side of the movable range of the hanger **103** in the case of the housing **101**. Additionally, in the present disclosure, the hanger **103** is arranged in the case of the housing **101**. Since at least the bifurcated portion and the rotation shaft portion supporting the housing **101** of the Y-shaped hanger **103** are arranged in the case of the housing **101**, in FIG. **1**, the bifurcated portion and the rotation shaft portion supporting the housing **101** of the hanger **103** are hidden in the housing **101** and are not visible. Then, only the joint portion with the slider **104** on the base side of the hanger **103** appears outside the case of the housing **101**. As described later, the microphone is arranged on the outer peripheral side of the movable range of the hanger, and the hanger **103** is incorporated in the case of the housing **101**, so that even if the tilt angle of the hanger **101** changes, the hanger **103** does not approach the vicinity of the microphone hole installed in the case of the housing **101**, and a difference does not occur in the acquired sound of the microphone. In addition, the problem of the wind noise generated by the air around the case of the housing **101** flowing around the hanger **103** is solved.

[0121] FIG. **4** illustrates a four-view diagram of the housing **101** of the headphones **100** (see FIG. **1**) according to an embodiment of the present disclosure viewed from four directions. In addition, FIG. **5** is a perspective view of the housing **101**, and FIG. **6** is a cross-sectional view of the housing **101** taken along line A-A of FIG. **4**. As can be seen from FIGS. **4** and **5**, the housing **101** is a dome-shaped three-dimensional structure, and has a substantially elliptical (or quadrangular with rounded four corners) contour. Note that while the housing **101** for the left ear is illustrated in FIGS. **4** to **6**, the housing **101** for the right ear has a substantially bilaterally symmetrical shape and structure.

[0122] Referring to the perspective view of FIG. **5**, the case of the housing **101** is divided into a side surface case **501** and a top surface case **502**, and an ear pad **107** is attached to a bottom surface portion of the side surface case **501**. Furthermore, referring to the cross-sectional view of FIG. **6**, the side surface case **501** supports a diaphragm **601** as a sounding body substantially at the center

such that the sound output side faces forward (downward in FIG. 6). Then, with the diaphragm **601** as a boundary, a front surface acoustic space **611** is formed on a front surface side (i.e., wearer's ear side) of the diaphragm **601**, and a first back surface space **612** is formed on a back surface side (i.e., outer side) of the diaphragm **601**. Furthermore, an outer peripheral space **613** is formed outside the front surface acoustic space **611** partitioned by an inner wall surface of the side surface case **501**. The hanger **103** is pivotably supported on a central projection of the side surface case **501**. The outer peripheral space **613** is a movable range of the hanger **103** and is an arrangement place of a plurality of microphones (only one microphone **604** is visible in cross-sectional view illustrated in FIG. 6). Additionally, in the top surface case **502**, a second back surface space **614** is formed further on the back surface side than the first back surface space **612**. A printed wiring board **602** on which electronic components including a drive circuit of the diaphragm **601** are mounted is arranged in the second back surface space **614**. Furthermore, the inside of the ring-shaped ear pad **107** is an NC region **615** to be subjected to noise canceling. In short, the housing **101** includes the side surface case **501**, the top surface case **502**, the ear pad **107**, and the diaphragm **601**, and has the acoustic space separation component on the side farther from the ear pad **107** with respect to the diaphragm **601**.

[0123] Note that the inside of the case of the housing **101** includes a first floor portion that is a non-acoustic space that does not need to be sealed and a second floor portion that is the second back surface space **614**. The second floor portion requires structural sealing of the space itself in order to control the characteristics of the diaphragm **601**, and is an acoustic space in which an acoustic duct (not illustrated) for adjusting the airflow volume exists in some cases. It can also be said that the housing **101** includes a plurality of acoustic spaces.

[0124] Referring again to FIG. 4, a Y-shaped bifurcated part of the hanger **103** accommodated in the case of the housing **101** is drawn by a broken line. A rotation shaft portion protruding inward at each distal end portion of the bifurcated portion of the hanger **103** is a movable shaft rotatably supported by a protruding portion at substantially the center of the side surface case **501**. Then, the hanger **103** is movable in the outer peripheral space **613** in the top surface case **502** about the movable shaft. The rotation angle about the movable shaft of the hanger **103** is the tilt angle of the housing **101**. The top surface case **502** is provided with a slit for moving a base portion (joint portion with headband **102**) of the hanger **103**. When the hanger **103** rotates about the movable shaft, the base portion of the hanger **103** can move up and down along the slit of the top surface case **502**. The hanger **103** is movable at a tilt angle in a range of 0 degrees to 15 degrees. FIG. 7 illustrates a side view and a cross-sectional view of the housing **101** when the tilt angle of the hanger **103** is minimum (0 degrees). Meanwhile, FIG. 8 illustrates a side view and a cross-sectional view of the housing **101** when the tilt angle of the hanger **103** is maximum (15 degrees). It can be understood from FIGS. 7 and 8 that the top surface case **502** of the housing **101** covers the movable range of the hanger **103**.

[0125] Next, the arrangement of the microphones with respect to the housing **101** having a structure in which the hanger **103** is taken into the case as illustrated in FIGS. 4 to 8 will be described.

[0126] As described above, in order to obtain good sound "quality" and "directivity" by performing signal processing on sound signals picked up by a plurality of microphones, it is important to arrange microphones close to the mouth of the wearer, secure a large distance between microphones, take measures against wind noise, for example. FIG. 9 illustrates an arrangement example of microphones satisfying these three conditions. FIG. 9 illustrates a state in which four microphones **901** to **904** are arranged in the side surface case **501** which is the first floor portion of the case of the housing **101**. Specifically, each microphone **901** to **904** is arranged in the outer peripheral space **613** on a side far from the ear pad **107** with the diaphragm **601** as a boundary, the outer peripheral space **613** being partitioned by the side surface case **501** and the top surface case **502**. In addition, it is assumed that a slit-shaped microphone hole (not illustrated) for sound pick-up

by the microphone is drilled in a wall surface (or joint portion between side surface case **501** and top surface case **501**) of the side surface case **501** near a place where each microphone **901** to **904** is arranged. One microphone **901** is arranged on the outer peripheral side of the hanger **103**, and the other three microphones **902** to **904** are arranged in consideration of other structures and acoustic characteristics. In the present disclosure, since the hanger **103** is incorporated in the case of the housing **101**, even if the tilt angle of the hanger **103** changes, the hanger **103** does not approach the vicinity of the microphone hole for the microphone **901**, and the problem of wind noise is solved. Note that the plurality of microphones **901** to **904** may be arranged in either the side surface case **501** as the first floor portion or the top surface case **502** as the second floor portion, but it can be said that it is more preferable to arrange the plurality of microphones in the side surface case **501** as the first floor portion from the viewpoint of securing a large inter-microphone distance.

[0127] FIG. **9** illustrates the arrangement example of the four microphones **901** to **904** in a case where the hanger **103** has a Y-shape and has two movable shafts. FIGS. **10** and **11** illustrate an arrangement example of four microphones **1001** to **1004** and **1101** to **1104** in a case where the hanger **103** has one movable shaft. In either case, one microphone **1001** or **1101** is arranged on the outer peripheral side of the hanger **103**, and the other three microphones **1002** to **1004** or **1101** to **1104** are arranged in consideration of other structures and acoustic characteristics.

[0128] FIG. **12** collectively illustrates microphone arrangement examples of a plurality of patterns. Note, however, that in FIG. **12**, the arrangement places of the microphones of the same set are represented by encircled numerals of the same value. That is, FIG. **12** illustrates microphone arrangement examples of four patterns numbered 1 to 4. As described above, the case of the housing **101** includes the side surface case **501** and the top surface case **502**. The first floor portion including the side surface case **501** is a non-acoustic space that does not need to be sealed, while the second floor portion including the top surface case **502** is an acoustic space that requires structural sealing of the space itself in order to control the characteristics of the diaphragm. In any of the patterns illustrated in FIG. **12**, one microphone is arranged on the outer peripheral side of the hanger **103**, and the other microphones are arranged in any of the side surface case **501** and the top surface case **502** in consideration of other structures and acoustic characteristics. In FIG. **12**, the portion of the top surface case **502** in the upper surface of the housing **101** is drawn in gray. The other part of the upper surface is the side surface case **501**.

[0129] Furthermore, FIG. **13** illustrates an arrangement example in which four microphones **1301** to **1304** are arranged separately in a first floor portion and a second floor portion of the housing **101**. In this example, too, one microphone **1301** is arranged on the outer peripheral side of the hanger **103**, and the other microphones **1302** to **1304** are arranged in consideration of other structures and acoustic characteristics. In FIG. **13**, too, the portion of the top surface case **502** in the upper face of the housing **101** is drawn in gray. The other part of the upper surface is the side surface case **501**.

[0130] Regarding the microphone arrangement method described in this section C, main features and effects thereof will be summarized.

[0131] (1) In the headphones **100** having the structure in which the housing **101** is supported via the hanger **103**, if a microphone can be arranged at a place on the outer peripheral side of the hanger **103** of the housing **101**, it is easy to secure a larger distance between the microphone and a microphone arranged at a position close to the mouth of the wearer of the headphones **100**.

[0132] (2) The movable part of the hanger **103** is arranged in the case of the housing **101**. Even if the tilt angle of the hanger **103** changes due to the shape of the side surface of the head of the wearer of the headphones **100** or the like, the space in front of the microphone hole is not affected, so that there is no difference in the acquired sound of the microphone. In addition, the problem of the wind noise generated by the air around the case of the housing **101** flowing around the hanger **103** is solved.

[0133] (3) The housing **101** has a structure in which one or more spaces (first back surface space

612 and second back surface space **614**) different from the space (outer peripheral space **613**) in which the hanger **103** and the microphone are present exist on a side far from the ear pad **107** (or human body) with the diaphragm **601** as a boundary. Therefore, by arranging the microphone in the outer peripheral space **613** that is a non-acoustic space, the microphone can pick up external sound while suppressing the influence of the sound generated from the diaphragm **601**. In addition, since structural sealing of the microphone is unnecessary, howling resistance is improved, in other words, noise canceling performance is improved.

D. Slider Structure

[0134] As described in the above section A, the slider **104** is coupled to the hanger **103** at the lower end, and has a slide function of adjusting the length by sliding in the extending direction of the headband **102** and a twist function of rotating the hanger **103** about the center line of the headband **102** to adjust the housing **101** to follow the shape of the temporal region of the wearer (or inclination of ear). However, in the structure of conventional headphones, there has been a problem in both cases of using the slide function and using the twist function of the slider.

[0135] First, the problem of the slide function of the slider **104** will be described. In order to improve marketability and quality, it is important to prevent a cable (hereinafter also referred to as "connection cable") that electrically connects the left and right housings **101** from being seen when the slider is largely pulled out from the headband **102** for the sake of appearance, and to make the headband **102** and the slider **104** have a lighter structure. However, conventionally, the connection cable has been sandwiched between two or more components to prevent it from being seen for the sake of appearance, or exposure of the connection cable has been allowed for the sake of weight reduction.

[0136] Next, the problem of the twist function of the slider **104** will be described. In a case where the headband **102** and the slider **104** have a flat cross-sectional shape as in the headphones **5900** illustrated in FIG. **59**, when the hanger **103** is twisted and rotated according to the shape of the temporal region of the wearer, a step is formed between the slider **104** and the hanger **103** as illustrated in FIG. **60**. The rotation angle of the hanger **103** varies and the step shape varies depending on the difference in the shape of the temporal region for each wearer. For this reason, there is a difference in the turbulence generated by the air flowing around the hanger **103** and the case of the housing (not illustrated in FIG. **60**) when the wearer walks, for example. As described in the above section C, in a case where the microphone is arranged at a place on the outer peripheral side of the hanger **104** of the housing, the environment in which the microphone picks up sound changes due to the step between the slider **104** and the hanger **103**, and thus, there is a concern about the influence on the microphone characteristics and the noise canceling performance.

[0137] In addition, when a step is generated between the slider **104** and the hanger **103** as illustrated in FIG. **60** due to the twist rotation of the hanger **103**, there is also a problem that the aesthetic appearance is impaired.

[0138] Therefore, in the present disclosure, at least the portion of the slider **104** joined to the hanger **103** has a cylindrical shape so as not to cause a step between the slider **104** and the hanger **103** at the time of twist rotation of the hanger **103**. In addition, the slider **104** has a hollow cylindrical structure, only the twist shaft portion of the hanger **103** is inserted into the hollow cylinder, and the connection cable from the housing **101** is inserted into and contained in the hollow cylinder, so that the connection cable is not exposed when the slider **104** slides in the extending direction of the head band **102**.

[0139] FIG. **14** illustrates a joint portion between the slider **104** and the hanger **103** in an enlarged manner in the headphones **100** according to the present disclosure. The slider **104** has a seamless short-arc hollow pipe shape and is inserted through a hollow cylindrical headband **102** (not illustrated in FIGS. **14** and **15**) to be the sheath side. The slider **104** can adjust the position of the housing **101** by sliding in an extending direction substantially coinciding with the center line of the

slider **104** itself. Furthermore, the slider **104** couples the hanger **103** that supports the housing **101** (not illustrated in FIGS. **14** and **15**) so that it is rotatable (twistable) about the center line of the slider **104** itself.

[0140] FIG. **15** illustrates a cross section of the joint portion between the slider **104** and the hanger **103** in the headphones **100** according to the present disclosure. In the hanger **103**, a lid hanger **1501** on the back surface side is removable, and the lid hanger **1501** can be removed to route a connection cable **1502** into the hanger **103**. One end of the connection cable **1502** is connected to the printed wiring board (described above) in the housing **101**, and the other end is inserted into the slider **104** that is the hollow cylinder via the hanger **103**. Therefore, since the connection cable **1502** is contained in the short-arc hollow pipe-shaped slider **104**, the connection cable **1502** is not exposed regardless of the position where the slider **104** slides in the extending direction of the headband **102**. As can be seen from FIGS. **14** and **15**, the hanger **103** and the slider **104** have a cylindrical structure having the same diameter, and are coaxially coupled to each other in a rotatable (twistable) manner.

[0141] The internal configuration of the joint portion between the slider **104** and the hanger **103** and the twist function, that is, the operation of rotating the hanger **103** about the center line of the slider **104** will be described with reference to FIG. **15**.

[0142] The slider **104** is made of resin molded in a short-arc hollow pipe shape as described above. A shaft twist slider **1503** is inserted into a joint portion of the slider **104** and the hanger **103** near the lower end of the slider **104** to integrate the slider **104** and the hanger **103**. The shaft twist slider **1503** is a metal nut component for fastening to a shaft twist **1504** on the hanger **103** side, and has a female screw structure formed on the inner peripheral side.

[0143] On the other hand, a male screw is formed at the distal end of the shaft twist **1504** on the hanger **103** side. When the lid hanger **1501** is removed, the inside of the hanger **103** becomes accessible. In a state where the lid hanger **1501** is removed, the shaft twist **1504** is inserted from below and aligned so as to be coaxial with the lower end surface of the shaft twist slider **1503** inserted in the slider **104**, and then the male screw at the distal end of the shaft twist **1504** is screwed to the female screw at the lower end of the shaft twist slider **1503** to form the fastening structure of the shaft twist **1504** and the shaft twist slider **1503**. Additionally, an O-ring **1505** is attached near the middle of the shaft twist **1504**. When the shaft twist **1504** and the shaft twist slider **1503** are fastened by the screw, the hanger **103** is in a coupled state in which it does not come off from the slider **104**. In this fastened state, the hanger **103** is rotatable about the center line of the slider **104** only by the frictional force generated by the O-ring **1505**, so that the twist function is achieved.

[0144] FIG. **16** illustrates an enlarged, upside-down view of the vicinity of the lower end of the slider **104** into which the shaft twist slider **1503** is inserted. Referring to FIG. **16**, a knurled portion **1601** having fine and regular uneven shapes is formed on the outer periphery of the shaft twist slider **1503** including a metal nut component. Additionally, a protrusion **1602** for restricting rotation with respect to the hanger **103** is formed on the upper end edge of the knurled portion **1601**. On the other hand, FIG. **57** illustrates the vicinity of the upper end of the hanger **103** in an enlarged manner. At the joint portion with the slider **104** on the hanger **103** side, a recess **5701** is formed in a range of the rotation angle in which the protrusion **1602** of the shaft twist slider **1503** is allowed to pass. In a state where the slider **104** and the hanger **103** are joined, the protrusion **1602** of the shaft twist slider **1503** is accommodated in the recess **5701** at the distal end of the hanger **103**, and the hanger **103** can rotate, that is, can twist about the center line of the slider **104** within the range of the angle in which the recess **5701** is formed.

[0145] Additionally, referring to FIG. **16**, a slit **1603** is formed at two locations of the knurled portion **1601** on the outer periphery of the shaft twist slider **1503**. In FIG. **16**, the slider **104** is drawn translucently, which is hard to see. Therefore, in FIG. **58**, the slider **104** is filled in with a dark color. The slider **104** is provided with projections **5801** at two locations at the end portion of

the hollow cylindrical inner wall. Therefore, the slit **1603** of the knurled portion on the shaft twist slider **1503** side and the projection **5801** on the inner periphery of the slider **104** are fixed by fitting, and rotate integrally even if torque is generated around the center line.

[0146] As described above, the slider **104** includes a resin having a short-arc hollow pipe shape, and the shaft twist slider **1503** including of a metal nut component is inserted at the end portion. A method of creating the slider **104** will be described with reference to the drawings as appropriate.

[0147] FIG. **17** is a perspective view of the slider **104** alone. Additionally, FIG. **18** is a side view of the slider **104**. Furthermore, FIG. **19** illustrates a cross-sectional view of the slider **104** cut along the center line. It is possible to create the slider **104** including an arc-shaped hollow molded article as illustrated in FIGS. **17** to **19** by one component by moving the mold along the arc in the direction of the arrow in FIG. **19**. By forming the slider **104** with one component using the arc slide structure, it is possible to obtain high rigidity sufficient to support the housing **101** with light weight and thin outer diameter. Mold slides are known in the art as one technology for injection molding undercuts (“shapes that do not come off with normal mold movement”) with a mold.

[0148] Furthermore, the shaft twist slider **1503**, which is a metal nut component, is inserted into the end portion of the molded slider **104** by thermal press fitting, so that the structure as illustrated in FIG. **16** can be completed. As a method of generating heat in thermal press-fitting, there is a method called impulse (heating by current). Note that examples of an alternative fastening means to thermal press-fitting include adhesion, a molding insert, and ultrasonic press-fitting.

[0149] FIG. **20** is an enlarged view of a joint portion between the slider **104** and the hanger **103**, including the housing **101**. The hanger **103** and the slider **104** have a cylindrical structure having the same diameter. In FIG. **20**, as indicated by an arrow **2002**, the hanger **103** and the slider **104** are coaxially coupled to each other in a rotatable manner, that is, in a twistable manner. Additionally, as described in the above section C, one microphone is arranged on the outer peripheral side of the hanger **103** (see FIG. **13**, for example), and therefore, a slit-shaped microphone hole **2001** is drilled near the base of the hanger **103**.

[0150] FIG. **21** illustrates a state in which air flows around the vicinity of the joint portion between the slider **104** and the hanger **103** when the wearer walks, for example. In FIG. **21**, air flows are indicated by arrows **2101** and **2102**. Note, however, that the arrow **2101** indicates a flow of air passing through the front side of the hanger **103** in FIG. **21**, and the arrow **2102** indicates a flow of air passing through the back side of the hanger **103** in FIG. **21**. Since the hanger **103** and the slider **104** have a cylindrical structure having the same diameter and the same axis, even if the hanger **103** rotates about the center line of the headband **102** by the twist function, no step (see FIG. **60**) is generated between the slider **104** and the hanger **103**, and an event such as changing of the step shape depending on the twist angle of the hanger **103** does not occur. Therefore, even if the hanger **103** is twisted, the environment in which the microphone picks up sound through the microphone hole **2001** arranged near the base of the hanger **103** hardly changes, and thus, there is no influence on the microphone characteristics and the noise canceling performance.

[0151] Regarding the structure of the slider described in this section D, main features and effects thereof will be summarized.

[0152] (1) Since the hanger **103** and the slider **104** having a cylindrical structure of the same diameter are coaxially coupled to each other in a rotatable manner to form a twist function, no step is generated even if the twist angle of the hanger **103** changes.

[0153] (2) By using the arc slide structure for the slider **103** and forming the slider **103** with one component including the shaft twist slider **1503**, it is possible to obtain high rigidity sufficient to support the housing **101** while achieving the slide function and the twist function with light weight and thin outer diameter.

[0154] (3) By adopting a structure in which no step is generated at a joint portion between the hanger **103** and the slider **104** when the hanger **103** is twisted, it is possible to arrange a microphone near the base of the hanger **103**. Even if the twist angle of the hanger **103** changes due

to the difference in the shape of the temporal region for each wearer of the headphones **101**, the microphone characteristics are not affected.

E. Ear Pad Structure

E-1. Moisture Permeable Synthetic Leather Ear Pad Structure

[0155] The ear pad **107** for headphones is required not to be stuffy for long-time comfort. As a method of coping with stuffiness, a method of imparting moisture permeability or air permeability to the material is known. On the other hand, as described in the above section B, the ear pad has a role of passive noise canceling that maintains the sound insulation property of the NC region. For example, it is not possible to apply, to headphones having a low-range reproduction of 100 Hz or less or a noise canceling function, an ear pad whose sealability decreases due to its air permeability. [0156] Synthetic leather and the like with moisture absorbing and releasing particles on the surface of synthetic leather are also known, but the effect of moisture absorption is limited, and there is also a problem of performance deterioration over time, such as hydrolysis. In addition, a film having moisture permeability is generally present, but it is not suitable as an exterior component of headphones due to problems of strength and appearance.

[0157] Therefore, in this section E-1, a structure of an ear pad that achieves both sound the insulation property (or soundproofing property) and moisture permeability according to the present disclosure will be described.

[0158] FIG. **22** illustrates a plan view of the ear pad **107** applied to the headphones **100** illustrated in FIG. **1**. Additionally, FIG. **23** is a cross-sectional view of the ear pad **107** illustrated in FIG. **22** taken along line B-B. As is known, the ear pad **107** is fixed to the bottom surface of the housing **101**. In the present disclosure, the ear pad **107** is replaceably attached to the housing **101**. Details will be described in the following section E-2.

[0159] As can be seen from FIG. **22**, the ear pad **107** has a ring-shaped structure and has a substantially elliptical (or quadrangular with rounded four corners) contour. The outer contour of the ear pad **107** is substantially the same as that of the bottom surface of the housing **101**. Furthermore, as can be seen from FIG. **23**, the ear pad **107** is configured by wrapping a cushion material **2301** including an elastic material such as polyurethane foam with a film **2302** such as synthetic leather.

[0160] Synthetic leather used for the film is usually (conventionally) formed on the outermost surface with a polyurethane film produced by a dry process. This polyurethane film is a non-porous film having a thickness of about 30 to 50 μm , and has no air permeability and poor moisture permeability. For reference, FIG. **24(A)** illustrates a cross-sectional structure of a normal wet synthetic leather, and FIG. **24(B)** illustrates a cross-sectional structure of a normal dry synthetic leather. As illustrated in FIG. **24(A)**, a normal wet synthetic leather is configured by bonding a porous (or air-permeable) wet layer **2402** including polyurethane to the front surface of a base cloth **2401**, and further forming a dry layer **2403** on the wet layer **2402**. On the other hand, in a normal dry synthetic leather, as illustrated in FIG. **24(B)**, a dry layer **2412** is directly formed on the outermost surface of a base cloth **2411**. Both the wet synthetic leather illustrated in FIG. **24(A)** and the dry synthetic leather illustrated in FIG. **24(B)** are impermeable and poor in moisture permeability.

[0161] FIG. **25** illustrates a cross-sectional structure of an example of synthetic leather used for a skin containing a urethane foam according to the present disclosure. The illustrated synthetic leather is configured by bonding a wet foam layer **2503** including porous (or air-permeable) polyurethane having a three-dimensional microporous structure on a base cloth **2501** of the lowermost layer, and a thin urethane film **2502** having a thickness of about 5 μm is integrated on the bottom surface side of the wet foam layer **2503**. For example, the thin urethane film **2502** can be formed on the bottom surface side of the wet foam layer **2503** using a coating technology. The urethane film **2502** is air-impermeable because it does not have open cells, but has moisture permeability and fine unevenness (surface transpiration) or moisture permeability of the skin.

[0162] FIGS. **26** and **27** each illustrates a modification of the synthetic leather configured by bonding a wet foam layer onto a base cloth according to the present disclosure illustrated in FIG. **25**. The example illustrated in FIGS. **26** and **27** is the same as FIG. **25** in that the base cloth **2501** and the wet foam layer **2503** are bonded to each other, but is different from the synthetic leather illustrated in FIG. **25** in that a thin urethane film **2601** is formed on the front surface side of the wet foam layer **2503** in the example illustrated in FIG. **26**, and thin urethane films **2701** and **2702** are formed on both the bottom surface and the front surface of the wet foam layer **2503** in the example illustrated in FIG. **27**.

[0163] In short, the synthetic leathers illustrated in FIGS. **25** to **27** have a structure in which a thin urethane film is integrated with at least one of the bottom surface or the front surface of the wet foam layer. The urethane film is air-impermeable because it does not have open cells, but has moisture permeability and fine unevenness (surface transpiration) or moisture permeability of the skin. In addition, none of the synthetic leathers illustrated in FIGS. **25** to **27** have an air-impermeable dry layer on the front surface of the wet foam layer. It can be said that the synthetic leathers illustrated in FIGS. **25** to **27** have both the sound insulation property and moisture permeability.

[0164] FIG. **28** illustrates a cross-sectional structure of another example of synthetic leather used for a skin containing a urethane foam according to the present disclosure. The illustrated synthetic leather is configured by bonding a wet foaming layer **2803** including porous (or air-permeable) polyurethane to the front surface of a wet base cloth **2801** having open cells (air permeability) as the lowermost layer, and an air-impermeable moisture-permeable waterproof film **2802** is arranged between the base cloth **2801** and the wet foaming layer **2803**. Moreover, FIG. **29** illustrates a modification of the synthetic film illustrated in FIG. **28**, which includes a base cloth and a wet foam layer and further includes an air-impermeable moisture-permeable film in combination. In the example illustrated in FIG. **29**, an air-impermeable moisture-permeable waterproof film **2901** is bonded to the bottom surface side of a wet base cloth **2902**, and a wet foam layer **2903** is bonded to the front surface side of the wet base cloth **2902**. It can be said that the synthetic leathers illustrated in FIGS. **28** and **29** have both the sound insulation property and moisture permeability.

[0165] FIG. **30** illustrates a cross-sectional structure of yet another example of synthetic leather used for a skin containing a urethane foam according to the present disclosure. The illustrated synthetic leather is configured by bonding a wet foaming layer **3002** including porous (or air-permeable) polyurethane to the front surface of a base cloth **3001** as the lowermost layer, and a dry layer **3003** is further bonded to the front surface of the wet foaming layer **3002**. In addition, FIG. **31** illustrates a modification of the synthetic film illustrated in FIG. **30**, which includes a wet foam layer and a base cloth and further includes a dry layer in combination. In the example illustrated in FIG. **31**, dry layers **3104** and **3101** are bonded to both the front surface of a wet foam layer **3103** and the bottom surface of a base cloth **3102**, respectively. The synthetic leathers illustrated in FIGS. **30** and **31** can improve the sound insulation property.

[0166] FIG. **32** illustrates a cross-sectional structure of still another example of synthetic leather used for a skin containing a urethane foam according to the present disclosure. The illustrated synthetic leather is configured by bonding a wet foaming layer **3203** including porous (or air-permeable) polyurethane on a base cloth **3201** as the lowermost layer, and dry layers **3204** and **3202** are bonded to both the front surface and the bottom surface of the wet foaming layer **3203**, respectively. In addition, FIG. **33** illustrates a modification of the synthetic film illustrated in FIG. **32**, which includes a wet foam layer and a base cloth and has a dry layer attached to the front surface. In the example illustrated in FIG. **33**, one or both of a wet foam layer **3303** and a wet base cloth **3301** having air permeability contain an impregnated material **3304**. In the example illustrated in FIG. **33**, a dry layer **3302** is bonded to the front surface of the wet base cloth **3301**, and then bonded to the wet foam layer **3303** containing the impregnated material **3304**. The impregnated material **3304** referred to herein is a resin, oil, or the like. The synthetic leathers illustrated in FIGS.

32 and **33** can improve the sound insulation property.

[0167] Regarding the moisture-permeable synthetic leather ear pad structure described in this section E-1, main features and effects thereof will be summarized.

[0168] (1) In an ear pad for headphones, in synthetic leather enclosing a cushion material including polyurethane foam or the like, by using a thin polyurethane film of about 5 to 20 μm which is air-impermeable but has moisture permeability, it is possible to achieve both the sound insulation property and moisture permeability of the ear pad.

E-2. Replaceable Claw Ear Pad

[0169] The ear pad is attached to the acoustic surface of the housing, and has a role of providing a better wearing feeling by its cushioning property and forming the NC region by its sound insulation property. As described in the above section E-1, the ear pad has a structure in which a cushion material such as polyurethane foam is wrapped with synthetic leather. However, there is a problem that synthetic leather on the surface deteriorates over time due to the influence of perspiration and body temperature, and it is desirable that the ear pad is replaceable.

[0170] Examples of a method of fixing the ear pad to the housing include a magnet type, a bayonet type (see Patent Document 5, for example), and a claw fixing type. Among these, the claw fixing type ear pad can be reduced in weight, but it is difficult to perform an operation of flexing the claw from the case surface of the housing, and thus, for example, it is difficult for the end user to remove and replace the ear pad.

[0171] Therefore, in the present disclosure, the claw fixing type is adopted as a method of fixing the replaceable ear pad **107** to the case of the housing **101**, and a structure of a fixing claw and a method of fixing the claw fixing type ear pad **107** that facilitate the removal work by the end user are proposed.

[0172] In the present embodiment, it is assumed that fixing claws are arranged at six locations on the inner peripheral side of the bottom surface of the ring-shaped ear pad **107**, and the ear pad **107** is attached to the acoustic surface of the housing **101** by inserting and locking each fixing claw into the corresponding claw fixing portion on the housing **101** side. Furthermore, by releasing the engagement state of each fixing claw with the claw fixing portion, the ear pad **107** can be removed from the housing **101** and be replaced. Note that in order to ensure acoustic sealing (or passive noise canceling function in NC region) by the ear pad **107**, it is desirable to fix the ear pad **107** to the acoustic surface of the housing **101** with the fixing claws and the claw fixing portions at a plurality of locations. In the present embodiment, fixing is performed at six locations, but fixing may be performed at five or less locations or seven or more locations as long as requirements such as acoustic sealing are satisfied.

[0173] FIG. **34** illustrates a cross-sectional view of the ear pad **107** and the housing **101** cut so as to include the vicinity of the claw fixing portion on the housing **101** side in a state where the fixing claw on the ear pad **107** side is locked. In addition, FIG. **35** further illustrates the fixing claw on the ear pad **107** side and the claw fixing portion on the housing **101** side in an enlarged manner in the cross-sectional view illustrated in FIG. **34**. The fixing claw on the ear pad **107** side has a claw fitting structure to be engaged with the claw fixing portion on the housing **101** side. Hereinafter, the claw fitting structure and the method of releasing the fitted state of the claw fitting structure will be described with reference to FIGS. **34** and **35**.

[0174] As illustrated in FIG. **35**, the fixing claw has a T-shaped cross section, and a projection **3501** protruding slightly in the direction of the foot of the T shape (or downward direction of T shape, left direction in FIG. **35**) is formed at one distal end of the T-shaped head. On the other hand, on the housing **101** side, a claw fixing portion including a rib **3511** is arranged at a portion facing the projection **3501**. Then, the ear pad **107** is fixed to the housing **101** by the claw fitting structure in which the projection **3501** on the fixing claw side is locked to the rib **3511** on the housing **101** side.

[0175] In addition, a leg portion of the T shape of the fixing claw is flexible and serves as a hinge **3502**. When the hinge **3502** flexes, the projection **3501** at the distal end is separated from the rib

3511 on the housing **101** side, so that the fitting state is released and the fixing claw can be removed from the claw fixing portion. In the fixing claw illustrated in FIG. **35**, the distal end of the T-shaped head opposite to the projection **3501** includes a hook-shaped finger hook portion **3503** that is bent back. Note, however, that the fixing claw is assumed to be attached to the ear pad **107** such that the projection **3501** faces the outer peripheral side of the ring and the finger hook portion **3503** faces the inner peripheral side. Then, as illustrated in FIG. **35**, when the fingertip is hooked on the finger hook portion **3503** and pressed in the direction of an arrow **3500** in FIG. **35**, the hinge **3502** flexes, and the projection **3501** on the opposite side is lifted up using the “principle of leverage” and detached from the rib **3511**. Thus, the fitted state can be easily released and the fixing claw can be easily detached from the claw fixing portion. The finger hook portion **3503** is supported by the hinge **3502** so as to face upward and be exposed from the inner periphery of the ring-shaped ear pad **107**.

[0176] As described above, in order to ensure acoustic sealing, the ear pad **107** has a claw fitting structure with (the acoustic surface of) the housing **101** including the fixing claw and the claw fixing portion at a plurality of locations (six locations in present example). Each claw fitting structure is as illustrated in FIGS. **34** and **35**. FIG. **36** illustrates an example in which six claw fitting structures are arranged in the ear pad **107** and the housing **101**. In FIG. **36**, six locations where claw fixing portions **3601** to **3606** are arranged are indicated by two solid squares and four broken squares. As illustrated in the drawing, the claw fitting structures at six locations are dispersedly arranged along the substantially elliptical contour of the ear pad **107** and the housing **101**. By evenly arranging the plurality of claw fitting structures along the substantially elliptical contour, the ear pad **107** can be firmly attached to the acoustic surface of the housing **101**, and acoustic sealing (or passive noise canceling function in NC region) by the ear pad **107** can be secured.

[0177] There are two types of fixing claws used in the claw fixing portions **3601** to **3606**, that is, a type having the finger hook portion **3503** as illustrated in FIG. **35** and a type not having the finger hook portion (not illustrated). Among the six places where the claw fixing portions **3601** to **3606** are arranged illustrated in FIG. **36**, a fixing claw having the finger hook portion is applied to the claw fixing portions **3601** and **3602** at the two locations indicated by solid squares, and a fixing claw not having the finger hook portion is applied to the claw fixing portions **3603** to **3606** at the four locations indicated by broken squares. The fixing claws having the finger hook portions **3503** at the two claw fixing portions **3601** and **3602** indicated by solid squares are slightly exposed to the inner peripheral side from the contour of the ear pad **107**. Since the finger hook portion **3503** of the fixing claw faces the inner peripheral side (described above), such an arrangement allows the user's fingertip to easily access the finger hook portion **3503**.

[0178] As can be seen from FIG. **36**, the two locations to which the fixing claw having the finger hook portion **3503** is applied are arranged side by side in a range where a curvature radius R of the elliptical ear pad **107** is small. FIG. **37** illustrates, in perspective view, arrangement places of the two claw fixing portions **3601** and **3602** arranged side by side in the range where R is small. In addition, FIG. **38** illustrates a state in which a finger is inserted into one of the two claw fixing portions **3601** and **3602** to remove the fixing claw. By arranging the claw having the finger hook portion in the range where R of the ear pad **107** is small, as illustrated in FIG. **38**, it is possible to warp the entire ear pad **107** only by inserting the finger into the range where R is small and applying a force to the ear pad **107** with the finger in the direction indicated by the arrow in FIG. **37**. Thus, the operability of removing the fixing claw is improved in the remaining four claw fixing portions **3603** to **3606** not having the finger hook portion.

[0179] Comparing the fixing claw of the type having the finger hook portion **3503** with the fixing claw of the type having no finger hook portion, the size of the fixing claw of the type having the finger hook portion is increased by the hook-shaped finger hook portion. Therefore, as illustrated in FIG. **39**, the fixing claw is exposed inward of the contour of the ring-shaped ear pad **107**. On the

other hand, when the headphones **100** are worn, the ear of the wearer is inserted into the ear pad **107**. FIG. **39** also illustrates a state in which the ear (auricle) is inserted into the elliptical interior of the ear pad **107**. As illustrated in FIG. **39**, it should be understood that the fixing claw having the finger hook portion is arranged at the front of the face so as not to come into contact with the ear and prevent adverse influence on the wearing feeling. In the fixing claw of the type having the finger hook portion, since the finger hook portion **3501** is exposed from the inside of the contour of the elliptical ear pad **107**, it can also be said that the finger hook portion **3501** is arranged at a visible position.

[0180] FIGS. **40** and **41** illustrate modifications of the finger hook portion of the fixing claw. The direction of pressing with the fingertip to release the state of fitting with the housing **101** and remove the fixing claw varies depending on the shape of the finger hook portion. In FIGS. **40** and **41**, directions in which the fixing claw is pressed by the fingertip to be removed are indicated by arrows. In the modification illustrated in FIG. **40**, a finger hook portion **4001** is pressed by the fingertip in the direction of an arrow **4000** in FIG. **40** (direction toward back of claw), and a projection **4002** at the distal end on the opposite side is lifted and detached from a rib **4011** on a housing **101** side using the “principle of leverage” similarly to the example illustrated in FIG. **35**. Alternatively, in the modification illustrated in FIG. **41**, a finger hook portion **4101** is pressed by the fingertip in a direction of an arrow **4100** in FIG. **41** (direction toward front of claw) to flex the entire claw, and a projection **4102** at the distal end on the opposite side is lifted and detached from a rib **4111** on a housing **101** side.

[0181] FIG. **42** illustrates another example in which a plurality of claw fixing portions for fixing the ear pad **107** to the acoustic surface on the housing **101** side is arranged. While FIG. **36** illustrates an example in which the claw fixing portions **3601** to **3606** are arranged at six locations, in the example illustrated in FIG. **42**, claw fixing portions **4201** to **4205** are arranged at five locations which are one location less. Note, however, that a fixing claw having the finger hook portion is applied to one claw fixing portion **4201** indicated by a solid square, and a fixing claw not having the finger hook portion is applied to the four claw fixing portions **4202** to **4205** indicated by broken squares. Similarly to the example illustrated in FIG. **36**, the fixing claw having the finger hook portion is applied only to the range where the R of the elliptical ear pad **107** is small, but only one location is provided. In a case where the demand for acoustic sealing (or passive noise canceling function in NC region) by the ear pad **107** is low, it is allowed to reduce the number of locations to which the fixing claw having the finger hook portion is applied.

[0182] Regarding the replaceable claw ear pad structure described in this section E-2, main features and effects thereof will be summarized.

[0183] (1) In the claw fixing type ear pad, a finger hook portion is formed at a position of the fixing claw facing the inner peripheral side of the ear pad. By hooking the fingertip on the finger hook portion and pressing the finger hook portion, the engagement relationship between the fixing claw and the housing side is released, and the ear pad can be easily removed.

[0184] (2) By dispersedly arranging the plurality of claw fixing portions along the elliptical contour of the ear pad, the ear pad can be more firmly fixed to the acoustic surface on the housing side, and acoustic sealing (or passive noise canceling function by ear pad) can be secured.

[0185] (3) By arranging the fixing claw having the finger hook portion among the plurality of claw fixing portions in the range where the curvature radius R is small, the entire ear pad can be warped when removing the ear pad, and the operation of removing the remaining fixing claw not having the finger hook portion becomes easy.

[0186] (4) By arranging the fixing claw having the finger hook portion at the front of the face so as not to come into contact with the ear, it is possible to prevent adverse influence on the wearing feeling.

F. Wiring Portion Sealing Structure

[0187] As illustrated in FIG. **6**, the housing **101** of the headphones **100** to which the present

disclosure is applied is divided into a plurality of spaces in the case. Referring to FIG. 6 in detail, the side surface case **501** supports the diaphragm **601**, and the front surface acoustic space **611** is formed on the front surface side of the diaphragm **601** and the first back surface space **612** is formed on the back surface side of the diaphragm **601** with the diaphragm **601** as a boundary. Furthermore, in the top surface case **502**, the second back surface space **614** further on the back surface side than the first back surface space **612** is formed, and the printed wiring board **602** on which the electronic components including the drive circuit of the diaphragm **601** are mounted is arranged in the second back surface space **614**. Therefore, on the wall separating the front surface acoustic space **611** and the second back surface space **614**, a wiring portion **603** including a hole for inserting a cable for electrically connecting the diaphragm **601** and the printed wiring board **602** is provided. In short, in a case where a plurality of electric components are arranged in a case partitioned into a plurality of spaces while being dispersed in two or more spaces, a wiring portion having a through-hole for inserting a cable is required for a wall separating the spaces.

[0188] The wiring that electrically connects the diaphragm **601** and the printed wiring board **602** includes a cable having a flat cross-sectional shape such as a flexible flat cable (FFC) or a flexible printed circuit (FPC). Then, in order to ensure the acoustic performance of each space, the wiring portion **603** through which this type of cable is inserted is strongly required to be acoustically sealed, in other words, to have a minute and constant airflow volume.

[0189] For example, a method is known in which when a cable is passed through a component forming a hole, a cushion material is sandwiched between the hole and the cable to ensure the sealability. The left side of FIG. **43** illustrates a state in which a hole through which a wire rod with a cushion material wrapped therearound passes is viewed from above. In addition, the right side of FIG. **43** illustrates a state in which a side surface of the wire rod with the cushion material wrapped therearound is viewed. In FIG. **43**, portions of the cushion material are indicated by hatching. However, in such a method, variations are large due to the work of wrapping the cushion material, and the airflow volume is not constant. Furthermore, voids are likely to be generated in a portion where the cushion materials are bonded together (see left side in FIG. **43**), and voids may be generated due to bonding deviation (see right side in FIG. **43**).

[0190] Furthermore, a flexible wiring board in which a waterproof member for sealing a through hole is integrally molded by insert molding has been proposed (see Patent Document 6), but a structure near the through hole is enlarged.

[0191] Additionally, a method of closing the void between the through hole and the wire rod by bond coating or the like is also exemplified, but there are problems of enlargement of the structure and contamination of peripheral components due to bond adhesion.

[0192] Therefore, in the present disclosure, an insertion type bushing (or joining part) that presses the wire rod such as an FFC or an FPC against a wall surface of the through hole is used to close the void between the through hole and the wire rod in the wiring portion. Thus, according to the present disclosure, it is possible to achieve a stable acoustic space in the wiring portion and to achieve downsizing of the wiring portion or the sealed portion with the minimum component (bushing) and the minimum structure volume.

[0193] FIG. **44** illustrates an external configuration example of a bushing **4400** according to the present disclosure. The illustrated bushing **4400** includes a flat pedestal portion **4401** having a large contour, and a rectangular parallelepiped projection **4402** protruding in the insertion direction from the upper surface of the pedestal portion **4401**. While FIG. **44** illustrates dimensions of about 5 mm×about 8 mm on the bottom surface and about 5 mm in thickness, these are design values that vary depending on the dimension of the through hole of the wiring portion **603** to which the bushing **4400** is applied. On the other hand, the cross-sectional shape of the projection **4402** is substantially the same as the shape of the through hole to which the bushing **4400** is applied. A sealing rib **4403** is formed substantially at the center of each of the four side surfaces of the projection **4402** over the entire circumference of the projection **4402**. Furthermore, a fall

prevention hook **4404** is formed at least at one location of an edge of an upper end surface of the projection **4402** (hooks **4404** are visible at two locations in FIG. **44**). Furthermore, the flat pedestal portion **4401** has a recessed shape (not visible in FIG. **44**) for facilitating grasping of the component on the opposite side (or side opposite to insertion direction) of the projection **4402**. [0194] The bushing **4400** is made of an elastic material such as silicon or elastomer. The hardness of the bushing **4400** is, for example, SHUR 80 degrees. More preferably, the surface of the bushing **4400** is provided with a lubricating coating in order to prevent getting stuck at the time of incorporation (at time of insertion into through hole of wiring portion **603**).

[0195] FIG. **45** is a cross-sectional view taken along the insertion direction, illustrating a state in which the bushing **4400** illustrated in FIG. **44** is inserted into the component **4500** forming the hole. Specifically, the component **4500** forming the hole is the wiring portion **603** drilled in a wall surface partitioning the first back surface space **612** and the second back surface space **614**. The bushing **4400** is inserted from bottom to top as indicated by an arrow in FIG. **45** into the component **4500** forming a hole through which a wire rod **4501** including FFC, FPC, or the like is inserted in advance.

[0196] Since the bushing **4400** has the pedestal portion **4401** larger than the hole, as can be seen from FIG. **45**, the bushing is inserted to a depth at which the pedestal portion **4401** reaches the hole. In addition, since the bushing **4400** has the hooks **4404** at two or more locations on the edge portion of the upper end surface, once the bushing is completely inserted into the component **4500** forming the hole, the hook **4404** is caught by a back end portion of the hole, and thus, the bushing **4400** can be prevented from falling off. Then, in a state where the bushing **4400** is inserted into the component **4500** forming the hole, the wire rod **4501** is pressed against the wall surface of the hole by the side surface of the bushing **4400**.

[0197] In addition, when the bushing **4400** is inserted into the hole, the sealing rib **4403** formed on the side surface of the projection **4402** over the entire circumference interferes with the wall surface of the hole and is pressed against the wall surface of the hole while being deformed, whereby the sealability is secured. As can be seen from the cross-sectional view of FIG. **45**, the sealing rib **4403** has a cross-sectional shape whose distal end has an acute angle and is along the wall surface of the hole. The sealing rib **4403** has, for example, a height of about 0.5 mm and an interference amount with the hole of about 0.2 mm. Therefore, in the sealing rib **4403**, since the main body of the bushing **4400** and the bushing **4400** are made of an elastic material such as silicon or elastomer, if the interference amount is about 0.2 mm, deformation to such an extent that the bushing **4400** can be inserted into the hole is allowed.

[0198] FIG. **46** is a cross-sectional view orthogonal to the insertion direction (or length direction of wire rod **4501**) illustrating a state in which the bushing **4400** illustrated in FIG. **44** is inserted into the component **4500** forming the hole. The bushing **4400** is made of an elastic material such as silicon or elastomer, and is inserted into the hole and deformed to close the void between the hole and the wire rod **4501**. As illustrated in FIG. **46**, although voids are generated at both ends of the wire rod **4501**, the airflow volume of the voids is minute and is constant regardless of the incorporation of the bushing **4400**. Therefore, with the method of closing the through hole of the wiring portion **603** using the bushing **4400** according to the present disclosure, acoustic design with stable sound insulation performance is possible.

[0199] FIG. **47** illustrates comparison of frequency characteristics of the headphones **100** between a state (normal state) in which the bushing **4400** is normally attached to the wiring portion **603** and a state (with partial floating) in which the bushing **4400** is intentionally attached so as to be partially floated by about 0.5 mm. It can be seen from FIG. **47** that the frequency characteristics hardly change even if the incorporated state of the bushing **4400** is changed, that is, a stable sound insulation performance can be obtained by the method for sealing the wiring portion using the bushing **4400** according to the present disclosure.

[0200] In the configuration example illustrated in FIG. **6**, the wiring portion **603** is provided only at

one location on the wall surface partitioning the first back surface space **612** and the second back surface space **614**, and is the only insertion place of the bushing **4400**. Note, however, that the installation place of the wiring portion is not limited to one location illustrated in FIG. **6**, and the application place of the wiring portion through which the wire rod is inserted, that is, the bushing, can be arranged at a plurality of locations in the case of the housing **101**. Moreover, although FIG. **45** illustrates an example in which the bushing **4400** is inserted upward from below, the insertion direction of the bushing is not limited to one direction.

[0201] FIGS. **48** and **49** illustrate a modification of the arrangement place of the wiring portion (or place where bushing is applied) in the case of the housing **101**. Note, however, that FIG. **48** illustrates a modification in a case where the insertion direction of the bushing is from bottom to top, and FIG. **49** illustrates a modification in a case where the insertion direction of the bushing is from top to bottom. In addition, although FIGS. **48** and **49** illustrate the candidate places of the plurality of wiring portions, how many places are actually used is a design matter. Additionally, the insertion directions of the plurality of bushings do not need to be unified, and the bottom to top direction and the top to bottom direction may be mixed.

[0202] Regarding the wiring portion sealing structure described in this section F, main features and effects thereof will be summarized.

[0203] (1) By inserting the bushing into the wiring portion between the spaces and closing the through hole while pressing the wire rod against the wall surface of the through hole, acoustic sealing can be achieved with minimum components and minimum structures. Since the minute void generated between the wire rod and the bushing is constant regardless of the incorporation of the bushing, acoustic design with stable sound insulation performance is possible.

[0204] (2) According to the configuration in which the bushing is inserted to seal the void of the wiring portion through which the wire rod is inserted, the bushing is easily inserted and removed, and manufacturability and reworkability are high.

[0205] (3) According to the configuration in which the bushing is inserted into the wiring portion between the spaces to close the through hole while pressing the wire rod against the wall surface of the through hole, it is not necessary to apply stress load, heat, and chemical influence to the wire rod.

G. Carrying Case

[0206] A carrying case for headphones is often used to facilitate carrying of the headphones and to store the headphones and prevent damage due to external force at the time of storage.

[0207] In a hard carrying case, the carrying case itself is required to have high rigidity in order to protect the headphones from external force and impact. Therefore, the size and thickness of the carrying case do not change regardless of whether the headphones are stored. When the carrying case not storing the headphones is put into a bag or the like or is put away in a drawer, the carrying case becomes unnecessarily bulky.

[0208] Therefore, the present disclosure proposes a carrying case for headphones that can maintain high rigidity for protecting the headphones from external force and impact when storing the headphones, and can reduce the volume when not storing the headphones so as not to be bulky when the headphones are put into a bag or put into a drawer.

[0209] Note that the carrying case described in this section G is basically assumed to store the headphones **100** of the present embodiment (or illustrated in FIG. **1**), but of course, the present disclosure can also be applied to a carrying case that stores other overhead headphones.

[0210] FIGS. **50** and **51** illustrate an example of a carrying case **5000** to which the present disclosure is applied. Note, however, that FIG. **50** illustrates a six-view diagram of the appearance of the carrying case **5000** as viewed from each of the upper, lower, front, rear, left, and right sides. In addition, FIG. **51** illustrates a two-view diagram in which the openable carrying case **5000** is opened as viewed from above and from one side surface. In addition, FIG. **52** is an exploded view of the carrying case **5000** disassembled into its main components.

[0211] Main components of the carrying case **5000** will be described with reference to FIG. **52**. The carrying case **5000** includes a top surface lid portion **5010**, a side surface portion **5020**, an internal tray portion **5030**, and a bottom surface tray portion **5040** in this order from the top. A strap **5050** is coupled to the distal end side of the bottom surface tray portion **5040**.

[0212] The top surface lid portion **5010** includes a top surface lid front portion **5011** including plate-shaped polycarbonate and a top surface lid back portion **5012** obtained by bonding a raised fabric to thin plate-shaped polycarbonate. After cutting each of the top surface lid front portion **5011** and the top surface lid back portion **5012** into a predetermined contour shape, the top surface lid front portion **5011** and the top surface lid back portion **5012** are bonded to each other, grooving is performed by thermocompression bonding, and then edge folding is performed to obtain the top surface lid portion **5010** having a shape bent in a mountain fold near the center as illustrated in FIG. **52**. The top surface lid portion **5010** has high rigidity to such an extent that it is not deformed even when an external force such as being crushed by a hand from above is applied, for example, and can protect the headphones **100** stored inside.

[0213] The internal tray portion **5030** mainly includes a material obtained by bonding a polyethyleneterephthalate (PET) resin material to a raised fabric, and a headphone storage pocket **5031** and an accessory storage pocket **5032** are formed by crimp molding. The headphone storage pocket **5031** includes a recess having substantially the same shape as the contour of the headphones **100**, and holds the headphones **100** in a swiveled state (left and right housings **101** are rotated so as to be accommodated in substantially the same plane as headband **102** by twist function of slider **104**). Additionally, the accessory storage pocket **5032** is a recess for storing accessory components formed at a place corresponding to the inside of the headband **102** of the headphones **100**. Moreover, the internal tray portion **5032** may further include an accessory pocket lid **5033**.

[0214] The bottom surface tray portion **5040** mainly includes a PET resin material, cut into a predetermined contour shape, and then molded into a container shape capable of accommodating the internal tray portion **5030** to be completed. Additionally, the strap **5050** is attached to the distal end side of the bottom surface tray portion **5040**. Similarly to the top surface lid portion **5010**, the bottom surface tray portion **5040** has high rigidity to such an extent that it is not deformed even when an external force such as being crushed by a hand from above is applied, for example, and can protect the headphones **100** housed inside.

[0215] The side surface portion **5020** is drawn in FIG. **52** in a relatively simple one ring shape, but includes an upper gusset portion **5021** and a lower gusset portion **5022** which are upper and lower parts divided by a fastener (not illustrated in FIG. **52**). FIG. **53** illustrates a developed view of the upper gusset portion **5021** and the lower gusset portion **5022**. Each of the upper gusset portion **5021** and the lower gusset portion **5022** includes a soft front cloth and back cloth, and a hard core material bonded to the front cloth. The development view illustrated in FIG. **53** illustrates a state in which a core material is bonded to each of the front cloths of the upper gusset portion **5021** and the lower gusset portion **5022**.

[0216] For each of the upper gusset portion **5021** and the lower gusset portion **5022**, the front cloth and the core material are cut into predetermined contour shapes, and then the front cloth and the core material are bonded together and sewn into a link shape, and a fastener fabric is further sewn. Moreover, the back fabric is also cut into a predetermined contour shape and then sewn into a ring shape, and the fastener fabric is further sewn to complete the side surface portion **5020**.

[0217] The side surface portion **5020** has different rigidity between a hard region (region colored in gray in each gusset portion in FIG. **53**) including the core material and a soft region (white region of each gusset portion in FIG. **53**) without the core material. Of the upper gusset portion **5021** and the lower gusset portion **5022**, the region to which the highly rigid core material is bonded is hardly bent. Therefore, in the upper gusset portion **5021** and the lower gusset portion **5022**, while the shape of the hard region including the core material is maintained, the soft region without the core material is deformed as a hinge, that is, a fold, and the side surface portion **5020** can be folded.

[0218] While the manufacturing process for each component of the carrying case **5000** has been described, the manufacturing process for assembling each component will also be described. First, the side surface portion **5020** and the bottom surface tray portion **5040** are sewn, and the strap **5050** is further attached to the bottom surface tray portion **5040**. Next, the top surface lid back portion **5012** and the side surface portion **5020** are sewn, the top surface lid back portion **5012** and the top surface lid front portion **5011** are bonded, and finally, the internal tray portion **5030** is attached to complete the carrying case **5000**.

[0219] Next, an operation of reducing the volume of the carrying case **5000** when not storing so as not to be bulky will be described.

[0220] As described above, the side surface portion **5020** has a hard region including a highly rigid core material and a hinge region without the core material, and the upper gusset portion **5021** and the lower gusset portion **5022** can be deformed with the hinge region without the core material serving as a fold while maintaining the shape of the hard region including the core material. In addition, a ring-shaped fastener region that joins the upper gusset portion **5021** and the lower gusset portion **5022** is also a hinge region that does not include a core material, and is a valley fold extending in the circumferential direction when the side surface portion **5020** is deformed. Then, the side surface portion **5020** can be folded by being bent such that a part of the hinge region not including the core material is valley folded and at least a part of the remaining hinge region not including the core material is mountain folded. For example, as a result of folding the side surface portion **5020** when the headphones **100** are not stored, the volume of the carrying case **5000** can be reduced so as not to be bulky.

[0221] The upper part of FIG. **54** illustrates a state of the carrying case **5000** before the side surface portion **5020** is folded (or when headphones **100** are stored). In addition, the lower part of FIG. **54** illustrates a state in which the top surface lid portion **5010** is pressed by a human hand from above and the side surface portion **5020** is folded when the headphones **100** are not stored. As illustrated in the drawing, the side surface portion **5020** is folded downward by bending the ring-shaped fastener region joining the upper gusset portion **5021** and the lower gusset portion **5022** so as to be valley folded. As a result, since the thickness of the carrying case **5000** is reduced, the entire volume is reduced, and it is possible to prevent bulkiness.

[0222] The folding structure is configured by arranging folds that are valley folds and mountain folds in the circumferential direction. When the side surface portion **5020** is folded in the headphones **100**, as illustrated in FIG. **54**, the hinge region having the fastener is valley folded, and at least a part of the remaining hinge region not including the core material is mountain folded. Parts of the side surface portion **5020** where folds serving as mountain folds are formed are illustrated with reference to FIGS. **55** and **56**.

[0223] Regarding the wiring portion sealing structure described in this section G, main features and effects thereof will be summarized.

[0224] (1) The carrying case to which the present disclosure is applied is a hard type in which both the top surface lid portion and the bottom surface tray portion have high rigidity and the side surface portion also includes a hard core material, and the headphones can be protected from external force and impact when the headphones are stored. On the other hand, in the side surface portion, a fastener portion joining the upper side and the lower side and a portion not including the core material serve as a hinge region, and a part of the hinge region is bent in a valley fold and another part of the hinge region is bent in a mountain fold, so that the side surface portion is folded in the thickness direction and becomes thin. Therefore, the headphones when not in use are folded to be compact so as not to be bulky, and can be easily stored away in a bag or a drawer.

INDUSTRIAL APPLICABILITY

[0225] The present disclosure has been described in detail with reference to the specific embodiments. However, it is obvious that those skilled in the art can make modifications and substitutions of the embodiment without departing from the gist of the present disclosure.

[0226] In the present specification, the embodiment in which the present disclosure is mainly applied to an overhead headphone including a noise canceling microphone has been mainly described, but the gist of the present disclosure is not limited thereto. For example, the structure of the slider described in the above section D, the ear pad structure described in the above section E, the wiring portion sealing structure described in the above section F, and the carrying case described in the above section G can be applied to a headphone not equipped with a microphone for noise canceling.

[0227] In short, the present disclosure has been described in an illustrative manner, and the contents disclosed in the present specification should not be interpreted in a limited manner. To determine the gist of the present disclosure, the claims should be taken into consideration.

[0228] Note that the present disclosure may also have the following configurations.

[0229] (1) Headphones including a headband, a housing, an ear pad attached to the housing, a hanger that supports the housing, and a slider that connects the hanger to an end portion of the headband, in which [0230] the headphones further include a microphone arranged at a place of the housing outside a movable range of the hanger.

[0231] (1-1) The headphones according to (1), in which [0232] the microphone is used to pick up an external sound for noise canceling.

[0233] (1-2) The headphones according to (1), in which [0234] a case of the housing has a hole for the microphone.

[0235] (2) The headphones according to (1), in which [0236] the housing includes a case that covers a movable range of the hanger.

[0237] (2-1) The headphones according to (2), in which [0238] the case of the housing has a slit for operating a joint portion of the hanger with the head band.

[0239] (3) The headphones according to any one of (1) and (2), in which [0240] the hanger and the microphone are arranged in the same space formed in the housing.

[0241] (3-1) The headphones according to (3), in which [0242] the microphone is arranged on an outer peripheral side of the hanger.

[0243] (4) The headphones according to any one of (1) to (3), in which [0244] the housing includes a diaphragm, and has one or more spaces different from a space in which the hanger and the microphone are arranged, on a side where the ear pad is far with the diaphragm as a boundary.

[0245] (5) The headphones according to any one of (1) to (4), in which [0246] another one or more microphones are arranged in a space different from the hanger and the microphone in the housing.

[0247] (6) The headphones according to any one of (1) to (4), in which [0248] the slider has a seamless short-arc hollow pipe shape, and contains a cable connected to a circuit in the housing in the hollow pipe.

[0249] (7) The headphones according to (6), in which [0250] the slider includes a resin having a seamless short-arc hollow pipe shape, and a shaft twist slider including a metal nut component inserted into a joint portion between the resin and the hanger.

[0251] (7-1) The headphones according to (7), in which [0252] the metal nut component is inserted into the resin by any one of thermal press-fitting, adhesion, molding insert, and ultrasonic press-fitting.

[0253] (8) The headphones according to (7), in which [0254] the slider rotatably supports the hanger about a center line of the slider by a fastening structure of the metal nut component and a distal end of a shaft twist inserted into the hanger.

[0255] (9) The headphones according to any one of (7) and (8), in which [0256] the metal nut component includes a protrusion that restricts a twist angle of the hanger.

[0257] (10) The headphones according to any one of (7) to (9), in which [0258] the metal nut component has a slit to be fitted with the resin having a short-arc hollow pipe shape.

[0259] (11) The headphones according to any one of (7) to (10), in which [0260] the slider has a cylindrical shape having the same diameter as the hanger, and is coaxially coupled with the hanger

in a rotatable manner.

[0261] (12) The headphones according to any one of (1) to (11), in which [0262] the ear pad is configured by containing a ring-shaped urethane foam in a skin, and [0263] the skin is configured by bonding a thin urethane film integrated with at least one of a bottom surface or a front surface of a wet foam layer having a three-dimensional microporous structure to a base cloth.

[0264] (13) The headphones according to any one of (1) to (11), in which [0265] the ear pad is configured by containing a ring-shaped urethane foam in a skin, and [0266] the skin is configured by bonding a wet foam layer having a three-dimensional microporous structure to a front surface of a product obtained by bonding an air-impermeable moisture-permeable waterproof film to any one of a bottom surface and a front surface of a wet base cloth having open cells or air permeability.

[0267] (14) The headphones according to any one of (1) to (11), in which [0268] the ear pad is configured by containing a ring-shaped urethane foam in a skin, and [0269] the skin is configured by bonding a wet foam layer having a three-dimensional microporous structure to a front surface of a product obtained by bonding a dry layer to only one of a bottom surface and a front surface of a wet base cloth.

[0270] (15) The headphones according to any one of (1) to (11), in which [0271] the ear pad is configured by containing a ring-shaped urethane foam in a skin, and [0272] the skin is configured by bonding a wet foam layer having a dry layer bonded to both a front surface and a back surface of the wet foam layer to a front surface of a base cloth, or configured by bonding a wet foam layer and a wet base cloth having air permeability, any one of the wet foam layer and the wet base cloth containing an impregnated material.

[0273] (16) The headphones according to any one of (1) to (15), in which [0274] the ear pad includes a fixing claw including a finger hook portion provided on an inner peripheral side of a ring shape.

[0275] (17) The headphones according to (16), in which [0276] the fixing claw has a claw fitting structure with a claw fixing portion on the housing side.

[0277] (18) The headphones according to any one of (16) and (17), in which [0278] the fixing claw includes a hinge that supports the finger hook portion, and when the finger hook portion is operated, the hinge is deformed and the fixing claw is released from the housing side.

[0279] (19) The headphones according to any one of (16) to (18), in which [0280] the ear pad includes fixing claws arranged at a plurality of locations on an inner periphery.

[0281] (20) The headphones according to (19), in which [0282] the fixing claws arranged at a plurality of locations include a fixing claw including a finger hook portion and a fixing claw not including a finger hook portion.

[0283] (21) The headphones according to (20), in which [0284] the fixing claws including a finger hook portion are arranged side by side in a range where a curvature radius R of the ear pad is small.

[0285] (22) The headphones according to any one of (20) and (21), in which [0286] the fixing claw including the finger hook portion is arranged at a front of a face.

[0287] (23) The headphones according to any one of (16) to (22), in which [0288] the finger hook portion is arranged at a visible position.

[0289] (24) The headphones according to any one of (1) to (23), further including [0290] a hole through which a wire rod is inserted into a wall that separates a space in the housing, and [0291] a bushing inserted into the hole so as to press the wire rod against a wall surface of the hole.

[0292] (24-1) The headphones according to (24), in which [0293] the bushing includes silicon, elastomer, or another material having elasticity.

[0294] (24-2) The headphones according to (24), in which [0295] a surface of the bushing is provided with a lubricating coating.

[0296] (24-3) The headphones according to (24), in which [0297] the hardness of the bushing is SHUR 80 degrees.

[0298] (25) The headphones according to (24), in which [0299] the bushing has a rib formed over

an entire circumference, and the rib interferes with a wall surface of the hole when the bushing is inserted into the hole.

[0300] (26) The headphones according to (25), in which [0301] the rib has an acute distal end and has a shape along a wall surface of the hole.

[0302] (27) The headphones according to any one of (24) to (26), in which [0303] the bushing includes a hook that is caught by a back end portion of the hole when the bushing is inserted into the hole.

[0304] (28) The headphones according to any one of (24) to (27), in which [0305] the bushing includes a flat pedestal portion having a protrusion on a side opposite to an insertion direction.

[0306] (29) A carrying case for the headphones according to claim 1, including: [0307] a top surface lid portion; [0308] a bottom surface tray portion; and [0309] a side surface portion having a ring shape sewn to the top surface lid portion and the bottom surface tray portion and having a fold that is folded by a pressing force in a vertical direction.

[0310] (30) The carrying case according to (29), in which [0311] a valley fold and a mountain fold are arranged in a circumferential direction in the side surface portion, and the side surface portion is folded at the fold by a pressing force in a vertical direction to reduce a thickness.

[0312] (31) The carrying case according to any one of (29) and (30), in which [0313] the side surface portion includes a hard portion configured by bonding a core material to a cloth and a soft portion including only the cloth, and the side surface portion is folded by using at least a part of the soft portion as a fold.

[0314] (32) The carrying case according to any one of (29) to (31), in which [0315] the side surface portion includes a fastener, and an upper gusset portion and a lower gusset portion connected via the fastener, and the side surface portion is folded by using at least the fastener as a fold.

[0316] (33) The carrying case according to any one of (29) to (32), in which [0317] the top surface lid portion has a shape bent in a mountain fold near a center.

[0318] (34) The carrying case according to any one of (29) to (33), further including [0319] a strap.

[0320] (35) The carrying case according to any one of (29) to (34), further comprising [0321] an internal tray portion accommodated in the bottom surface tray unit, and having a headphone storage pocket including a recess along a contour of the headphones according to claim 1.

REFERENCE SIGNS LIST

[0322] **100/5900** Headphones [0323] **101** Housing [0324] **102** Headband [0325] **103** Hanger [0326] **104** Slider [0327] **106** Speaker [0328] **107** Ear pad [0329] **302** FF microphone [0330] **303** FF cancellation signal generation unit [0331] **304** Reproduction unit [0332] **305** Synthesis unit [0333] **306** FB microphone [0334] **307** FB cancellation signal generation unit [0335] **501** Side surface case [0336] **502** Top surface case [0337] **601** Diaphragm [0338] **602** Printed wiring board [0339] **603** Wiring portion [0340] **604** Microphone [0341] **611** Front surface acoustic space [0342] **612** First back surface space [0343] **613** Outer peripheral space [0344] **614** Second back surface space [0345] **615** NC region [0346] **901** to **904** Microphone [0347] **1001** to **1004** Microphone [0348] **1101** to **1104** Microphone [0349] **1301** to **1304** Microphone [0350] **1501** Lid hanger [0351] **1502** Connection cable [0352] **1503** Shaft twist slider (metal nut component) [0353] **1504** Shaft twist [0354] **1505** O-ring [0355] **1601** Knurled portion [0356] **1602** Protrusion [0357] **5701** Recess [0358] **1603** Slit [0359] **5801** Projection [0360] **2001** Microphone hole [0361] **2301** Cushion material [0362] **2302** Film [0363] **2501** Base cloth [0364] **2502** Urethane film [0365] **2503** Wet foam layer [0366] **2601** Urethane film [0367] **2701**, **2702** Urethane film [0368] **2801** Wet base cloth [0369] **2802** Moisture-permeable prevention film [0370] **2803** Wet foam layer [0371] **2901** Moisture-permeable prevention film [0372] **2902** Wet base cloth [0373] **2903** Wet foam layer [0374] **3001** Base cloth [0375] **3002** Wet foam layer [0376] **3003** Dry layer [0377] **3101** Dry layer [0378] **3102** Base cloth [0379] **3103** Wet foam layer [0380] **3104** Dry layer [0381] **3201** Base cloth [0382] **3202** Dry layer [0383] **3203** Wet foam layer [0384] **3204** Dry layer [0385] **3301** Wet base cloth [0386] **3302** Dry layer [0387] **3303** Wet foam layer [0388] **3304** Impregnated material [0389]

3501 Projection [0390] **3502** Hinge [0391] **3503** Finger hook portion [0392] **3511** Rib [0393] **3601** to **3606** Claw fixing portion [0394] **4001** Finger hook portion [0395] **4002** Projection [0396] **4011** Rib [0397] **4101** Finger hook portion [0398] **4102** Projection [0399] **4111** Rib [0400] **4201** to **4205** Claw fixing portion [0401] **4400** Bushing [0402] **4401** Pedestal portion [0403] **4402** Projection [0404] **4403** Sealing rib [0405] **4404** Hook [0406] **4500** Component forming hole [0407] **5000** Carrying case [0408] **5010** Top surface lid portion [0409] **5011** Top surface lid front portion [0410] **5012** Top surface lid back portion [0411] **5020** Side surface portion [0412] **5021** Upper gusset portion [0413] **5022** Lower gusset portion [0414] **5030** Internal tray portion [0415] **5031** Headphone storage pocket [0416] **5032** Accessory storage pocket [0417] **5033** Accessory pocket lid [0418] **5040** Bottom surface tray portion [0419] **5050** Strap

Claims

1. Headphones comprising a headband, a housing, an ear pad attached to the housing, a hanger that supports the housing, and a slider that connects the hanger to an end portion of the headband, wherein the headphones further comprise a microphone arranged at a place of the housing outside a movable range of the hanger.
2. The headphones according to claim 1, wherein the housing includes a case that covers a movable range of the hanger.
3. The headphones according to claim 1, wherein the hanger and the microphone are arranged in a same space formed in the housing.
4. The headphones according to claim 1, wherein the housing includes a diaphragm, and has one or more spaces different from a space in which the hanger and the microphone are arranged, on a side where the ear pad is far with the diaphragm as a boundary.
5. The headphones according to claim 1, wherein another one or more microphones are arranged in a space different from the hanger and the microphone in the housing.
6. The headphones according to claim 1, wherein the slider has a seamless short-arc hollow pipe shape, and contains a cable connected to a circuit in the housing in the hollow pipe.
7. The headphones according to claim 6, wherein the slider includes a resin having a seamless short-arc hollow pipe shape, and a shaft twist slider including a metal nut component inserted into a joint portion between the resin and the hanger.
8. The headphones according to claim 7, wherein the slider rotatably supports the hanger about a center line of the slider by a fastening structure of the metal nut component and a distal end of a shaft twist inserted into the hanger.
9. The headphones according to claim 7, wherein the metal nut component includes a protrusion that restricts a twist angle of the hanger.
10. The headphones according to claim 7, wherein the metal nut component has a slit to be fitted with the resin having a short-arc hollow pipe shape.
11. The headphones according to claim 7, wherein the slider has a cylindrical shape having a same diameter as the hanger, and is coaxially coupled with the hanger in a rotatable manner.
12. The headphones according to claim 1, wherein the ear pad is configured by containing a ring-shaped urethane foam in a skin, and the skin is configured by bonding a thin urethane film integrated with at least one of a bottom surface or a front surface of a wet foam layer having a three-dimensional microporous structure to a base cloth.
13. The headphones according to claim 1, wherein the ear pad is configured by containing a ring-shaped urethane foam in a skin, and the skin is configured by bonding a wet foam layer having a three-dimensional microporous structure to a front surface of a product obtained by bonding an air-impermeable moisture-permeable waterproof film to any one of a bottom surface and a front surface of a wet base cloth having open cells or air permeability.
14. The headphones according to claim 1, wherein the ear pad is configured by containing a ring-

shaped urethane foam in a skin, and the skin is configured by bonding a wet foam layer having a three-dimensional microporous structure to a front surface of a product obtained by bonding a dry layer to only one of a bottom surface and a front surface of a wet base cloth.

15. The headphones according to claim 1, wherein the ear pad is configured by containing a ring-shaped urethane foam in a skin, and the skin is configured by bonding a wet foam layer having a dry layer bonded to both a front surface and a back surface of the wet foam layer to a front surface of a base cloth, or configured by bonding a wet foam layer and a wet base cloth having air permeability, any one of the wet foam layer and the wet base cloth containing an impregnated material.

16. The headphones according to claim 1, wherein the ear pad includes a fixing claw including a finger hook portion provided on an inner peripheral side of a ring shape.

17. The headphones according to claim 16, wherein the fixing claw has a claw fitting structure with a claw fixing portion on the housing side.

18. The headphones according to claim 16, wherein the fixing claw includes a hinge that supports the finger hook portion, and when the finger hook portion is operated, the hinge is deformed and the fixing claw is released from the housing side.

19. The headphones according to claim 16, wherein the ear pad includes fixing claws arranged at a plurality of locations on an inner periphery.

20. The headphones according to claim 19, wherein the fixing claws arranged at a plurality of locations include a fixing claw including a finger hook portion and a fixing claw not including a finger hook portion.

21. The headphones according to claim 20, wherein the fixing claws including a finger hook portion are arranged side by side in a range where a curvature radius R of the ear pad is small.

22. The headphones according to claim 20, wherein the fixing claw including the finger hook portion is arranged at a front of a face.

23. The headphones according to claim 16, wherein the finger hook portion is arranged at a visible position.

24. The headphones according to claim 1, further comprising a hole through which a wire rod is inserted into a wall that separates a space in the housing, and a bushing inserted into the hole so as to press the wire rod against a wall surface of the hole.

25. The headphones according to claim 24, wherein the bushing has a rib formed over an entire circumference, and the rib interferes with a wall surface of the hole when the bushing is inserted into the hole.

26. The headphones according to claim 25, wherein the rib has an acute distal end and has a shape along a wall surface of the hole.

27. The headphones according to claim 24, wherein the bushing includes a hook that is caught by a back end portion of the hole when the bushing is inserted into the hole.

28. The headphones according to claim 24, wherein the bushing includes a flat pedestal portion having a protrusion on a side opposite to an insertion direction.

29. A carrying case for the headphones according to claim 1, comprising: a top surface lid portion; a bottom surface tray portion; and a side surface portion having a ring shape sewn to the top surface lid portion and the bottom surface tray portion and having a fold that is folded by a pressing force in a vertical direction.
